

Graded Assignment: CI/CD Pipeline

Graded Project — CI/CD Pipeline

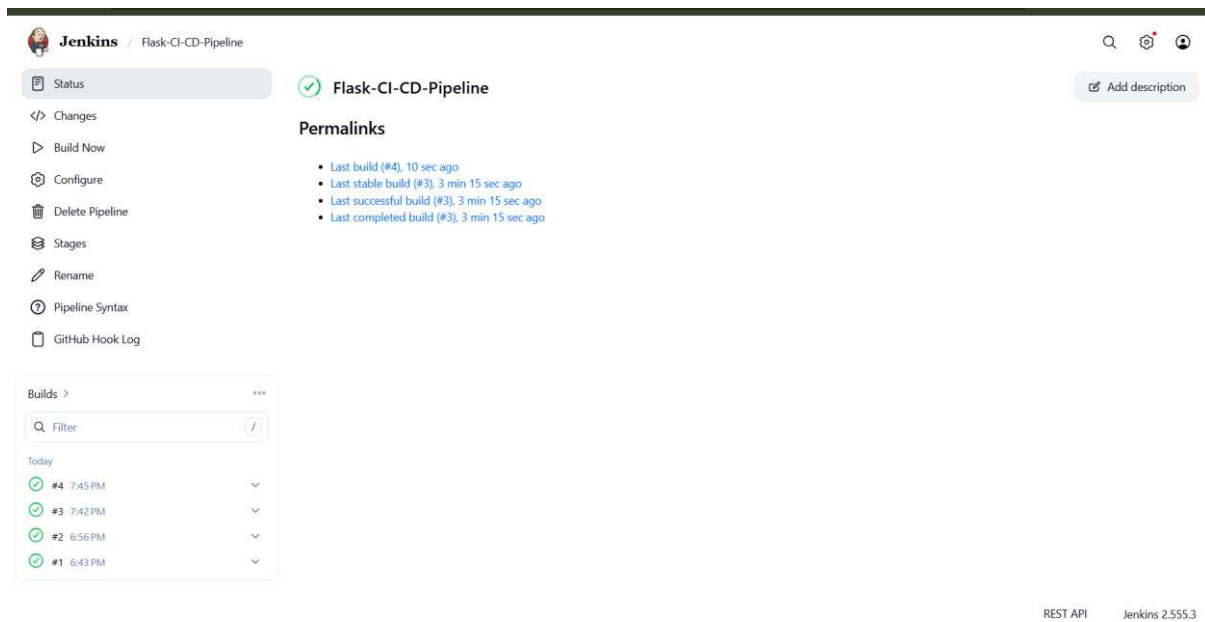
Assignment 1: Jenkins CI/CD Pipeline for Flask Application

Assignment 2: GitHub Actions CI/CD Pipeline for Flask Application

GitHub - <https://github.com/kaushik0203/sample-python-api.git>

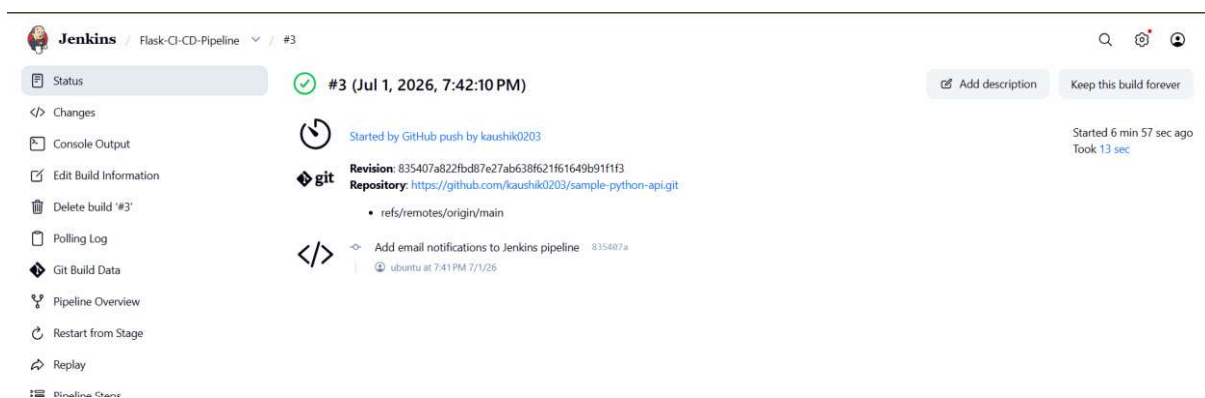
Submitted by – Koushik Thalla

Email – Kaushikthalla@gmail.com



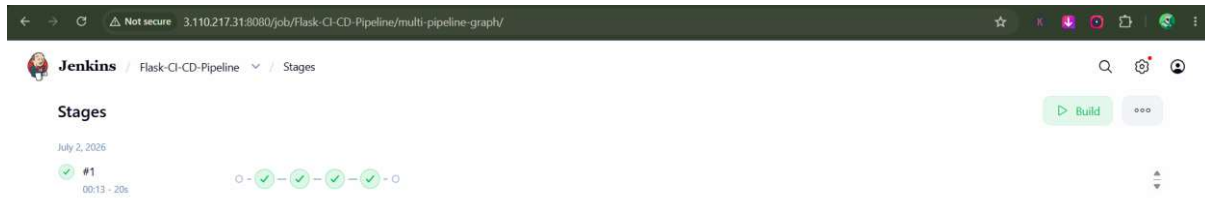
Screenshot 1 – Jenkins Dashboard

The Jenkins dashboard showing the configured CI/CD project used for automating the Flask application build, testing, and deployment



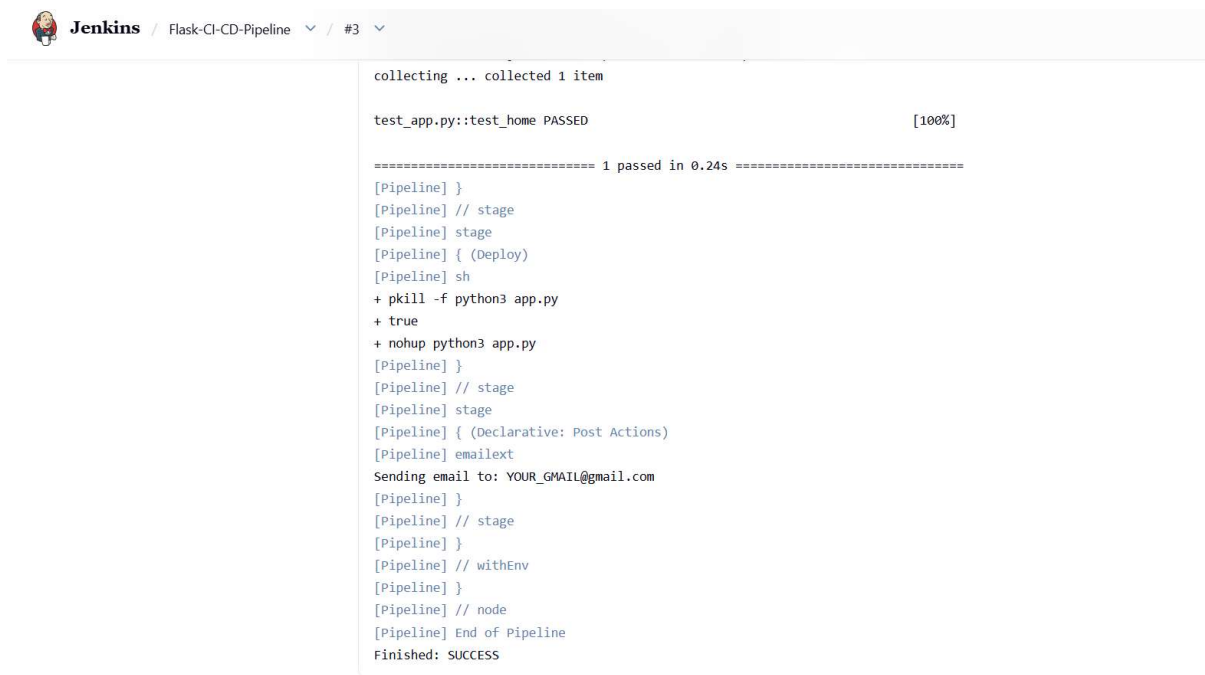
Screenshot 2 – Jenkins Pipeline Stage View

Pipeline Stage View displaying the successful execution of the Build, Test, and Deploy stages



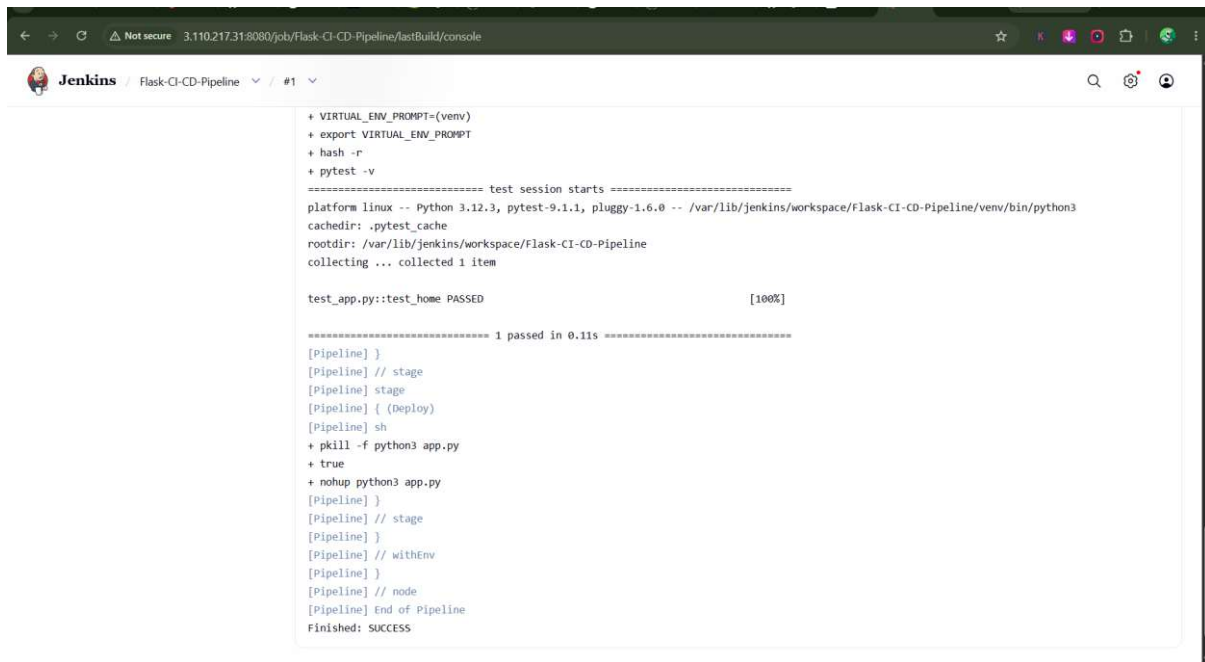
Screenshot 3 – Jenkins Pipeline Overview

Jenkins Pipeline Overview showing the completed pipeline execution and build history



Screenshot 4 – Build Console Output

Console output showing dependency installation, execution of unit tests using pytest, and successful deployment steps



```
+ VIRTUAL_ENV_PROMPT=(venv)
+ export VIRTUAL_ENV_PROMPT
+ hash -r
+ pytest -v
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /var/lib/jenkins/workspace/Flask-CI-CD-Pipeline/venv/bin/python3
cachedir: .pytest_cache
rootdir: /var/lib/jenkins/workspace/Flask-CI-CD-Pipeline
collecting ... collected 1 item

test_app.py::test_home PASSED [100%]

===== 1 passed in 0.11s =====
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] sh
+ pkill -f python3 app.py
+ true
+ nohup python3 app.py
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withenv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

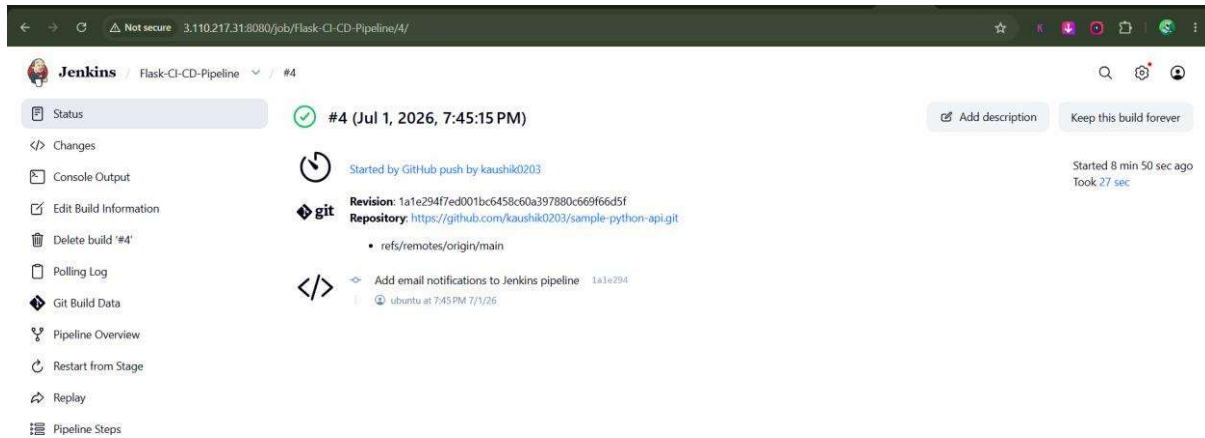
Screenshot 5 – Jenkins Console Output

Detailed Jenkins execution log confirming that all pipeline stages completed without errors



Screenshot 6 – Email Notification

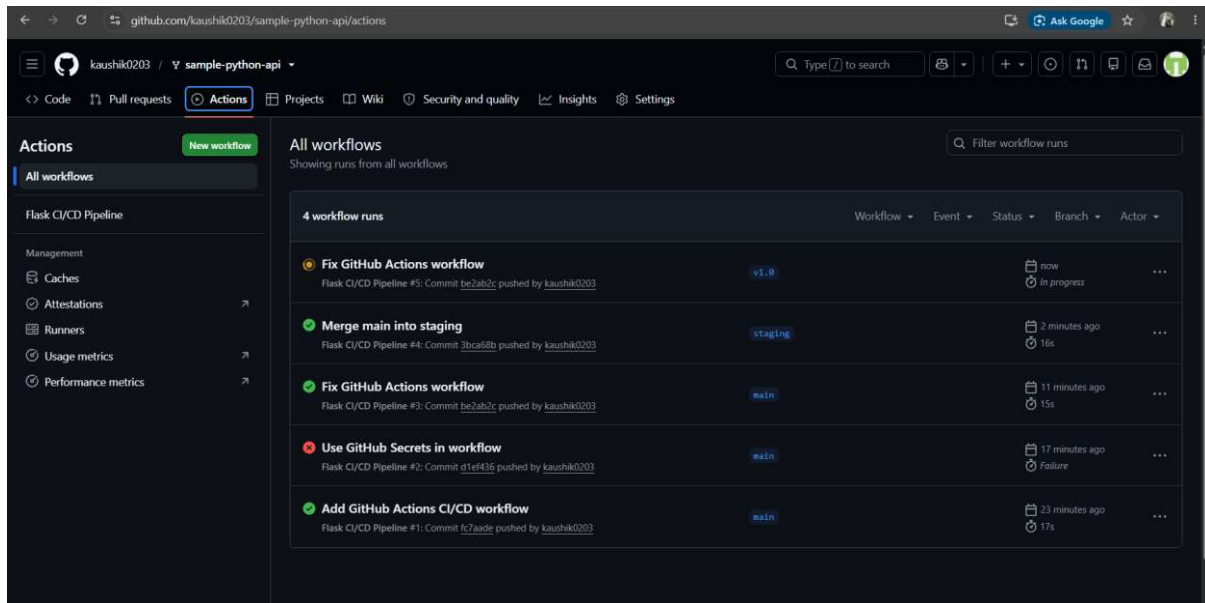
Gmail notification received after Jenkins completed the build successfully confirming email notification configuration



Screenshot 7 – Final Build Confirmation

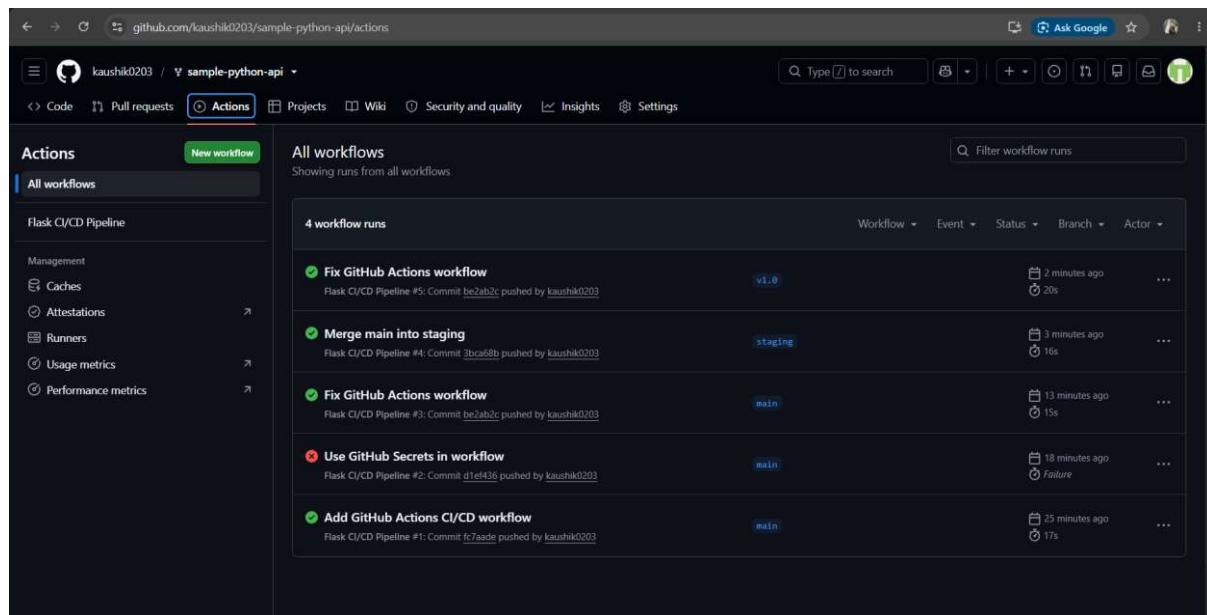
Final successful Jenkins build confirming that the complete CI/CD pipeline executed correctly

ASSIGNMENT 2 — GITHUB ACTIONS



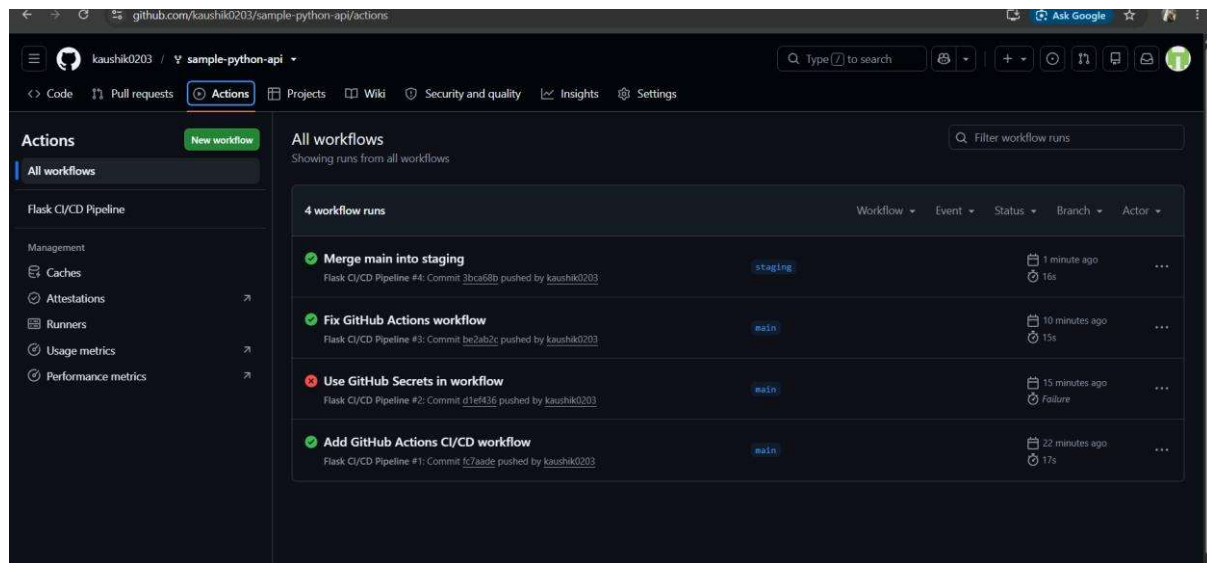
Screenshot 1 - Initial GitHub Actions Workflow

Initial execution of the GitHub Actions workflow during workflow configuration and validation



ScreenShot 2 - GitHub Actions Main Workflow

GitHub Actions workflow successfully executed after a push to the main branch, completing dependency installation, testing, and build stages



Screenshot 3 - Staging Deployment

Successful deployment to the staging environment after pushing changes to the staging branch